THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 3: **Metals and Non-metals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following is an ionic compound? 1

(a) CH_4 **(b)** CCl_4 **(c)** $MgCl_2$ **(d)** H_2O

2. Brass is an alloy of: 1

(a) iron and carbon **(b)** copper and zinc **(c)** lead and tin **(d)** copper and tin

3. Oxides of metals which react with both acids and bases are called: 1

(a) acidic oxides **(b)** neutral oxides
(c) basic oxides **(d)** amphoteric oxides

4. Ionic compounds conduct electricity: 1

(a) in solid state **(b)** in solid and molten state
(c) never **(d)** in molten state and in aqueous solution

5. Bronze is an alloy of: 1

(a) copper and tin **(b)** iron and nickel **(c)** lead and tin **(d)** copper and zinc

6. Which metal is stored under kerosene oil? 1

(a) copper **(b)** zinc **(c)** sodium **(d)** iron

7. The ore of mercury is: 1

(a) bauxite **(b)** haematite **(c)** galena **(d)** cinnabar

8. Solder is an alloy of: 1

(a) copper and tin **(b)** lead and tin **(c)** iron and carbon **(d)** copper and zinc

9. The metal which is liquid at room temperature is: 1

(a) iron **(b)** mercury **(c)** aluminium **(d)** sodium

10. The ore of aluminium is: 1

(a) bauxite **(b)** cinnabar **(c)** zinc blende **(d)** haematite

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why are food cans coated with tin and not zinc? 2

12. What is galvanisation? State its purpose. 2

13. Write the equation for the reaction of aluminium oxide with HCl and with NaOH to show it is amphoteric. 2

14. What is meant by an ore and gangue? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. What happens when zinc is added to copper sulphate solution? Write the equation. 3

16. What is thermit reaction? Give the equation and an application. 3

17. Why are copper and aluminium used to make electric wires? Mention one more use of each. 3

OR

Explain the formation of $MgCl_2$ by transfer of electrons (atomic numbers Mg = 12, Cl = 17). 3

18. Describe the extraction of sodium from molten sodium chloride. 3

OR

Write the reaction of aluminium with dilute HCl and name the gas liberated and its test. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. What are the reactivity series and displacement? Describe the iron and copper sulphate experiment. 5

OR

What are the physical properties of non-metals? Give two exceptions with reasons. 5

20. Describe the formation of NaCl and $MgCl_2$ by electron transfer and state four properties of ionic compounds. 5

OR

What are amphoteric oxides? Write equations to show amphoteric nature of Al_2O_3 and ZnO. 5

END OF QUESTION PAPER — Total: 40 Marks

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